

MLib Yield Curve User Guide

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MHI

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MLib Analytics Framework

- MHI use the MLib C++ analytics library for curve construction
- There are 2 Excel Add-ins: Vanilla-Addin & eTrading-Addin
- The eTrading-Addin is used for Swaps eTrading

Yield Curves:

- MLib supports Multi-Yield Curves
- OIS Discounting and CSA support is available
- In Excel a full set of USD curves build in approx 0.6 seconds
i.e. USDOIS, USD1ML, USD3ML, USD6ML and USD12ML curves

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Multi-Curves

MLib supports a full variety of curves including OIS, Standard Swap, Tenor Basis, Xccy Basis and CSA curves. These can be constructed using the below functions within the eTrading Excel addin

Curve Types:

`meCurveCalibrateOIS`

OIS Curves

`meCurveCalibrateSwap`

Standard Swap Curves

`meCurveCalibrateBasis`

Tenor Basis Curves

CSA Curves:

`meCurveCalibrateBasis`

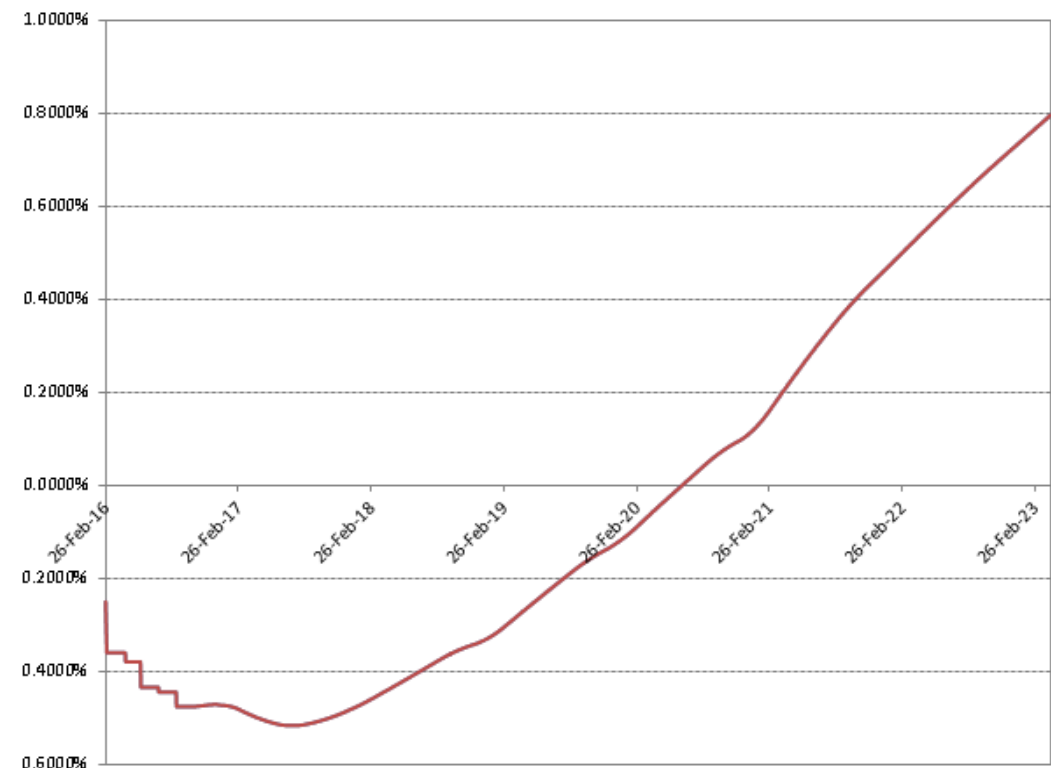
Xccy Basis Curves

`meCurveCalibrateFXForwards`

CSA Curves

Calibration Instruments

- OIS Swaps
- Libor-OIS Basis Swaps
- Central Bank Dated Swaps
- e.g. ECB and Fed Fund Swaps



Curve Features

OIS curves are typically constructed using OIS Swaps and Libor-OIS Basis Swaps. The short end is interpolated piecewise constant between Central Bank meeting dates.

meCurveCalibrateOIS ("CurveCollection", "StaticDataTable", "CurveIndex", "CurveBuildProperties", "OISConventions", "OISMarketData", "OISFixingTable", "OISBasisConventions", "OISBasisMarketData", "SwapConventions", "SwapMarketData")

meCurveCalibrateOIS

CurveCollection: Name of the Curve Group, usually grouped by Currency StaticDataTable: (Case Sensitive!) A hard-coded reference to a property file table used to lookup default curve parameters if not specified IndexName: A user defined name(s) for the output curve index, multiple aliases allowed. e.g. EUROIS, EUR3ML, EUR6ML		<i>meCurveCalibrateOIS</i> Curve Loaded	EURYC OIS Curve has been set (OIS) TRUE
		Curve Names	
		CurveCollection	EURYC
		StaticDataTable	OIS
		IndexName(s)	OIS:EUROIS:EURDF:EURDF_NO

Curve Build Properties

AsOfDate: The curve build Date Currency: The curve currency Interpolation: The interpolation method used for getting the OIS discount factors YieldGen.Interpolation: The interpolation method used for generating the target curve forecast rates		Curve Build Properties	
		AsOfDate	31-May-16
		Currency	EUR
		interpolation	fn_SplineInterpolation
		yieldgen.interpolation	fn_SplineInterpolation

OIS Curve Conventions

Overnight Indexed Swaps

	OIS Conventions		OIS Par Rates					
				Term	Rate	StartDate	EndDate	Use
Calendar: The fixed leg calendar	calendar	TGT						
Daycount: The fixed leg daycount	daycount	ACT/360	CASH	ON	-0.345%			TRUE
Frequency: The fixed leg frequency	frequency	Annual	ECB SWAP	EUSF1A	-0.340%	02-Jun-16	21-Jul-16	TRUE
ResetLag: The float fixing lag	resetlag	2d	ECB SWAP	EUSF2A	-0.350%	21-Jul-16	08-Sep-16	TRUE
SlidingRule: The fixed leg BusinessDayAdjustment	slidingrule	MOD_FOLLOWING	ECB SWAP	EUSF3A	-0.371%	08-Sep-16	20-Oct-16	TRUE
IsEomRoll: Use the End of Month Roll Convention for swaps traded on End of Month	iseomroll	TRUE	ECB SWAP	EUSF4A	-0.382%	20-Oct-16	08-Dec-16	TRUE
EomDay: Day to be considered month end	eomday	31	OIS SWAP	12M	-0.382%			TRUE
GenerateMethod: Compounding method, can be "DailyCompounding" or remove this key and value for no compounding	generatemethod	dailycompounding	OIS SWAP	18M	-0.395%			TRUE
FirstRate: This field controls the first rate to be used for curve construction; "ShortMarket" implies include short end cash instruments. Removing this key and value will start curve construction from the Swaps instruments	firstrate	ShortMarket	OIS SWAP	2Y	-0.401%			TRUE
ShortTermConvention: The OIS compound method to be used. Can be "ArithmeticAverage" and if the key and value are removed will default to GeometricAverage	shorttermconvention	arithmeticaverage	OIS SWAP	3Y	-0.385%			TRUE
LongTermConvention: A flag to indicate if Libor OIS Spread instruments should be used on the Long End of the OIS Curve; Can be "LOBASIS" to use Libor OIS Spread instruments and if the key and value are removed only OIS Swap Outrights will	longtermconvention	LOBASIS	OIS SWAP	4Y	-0.342%			TRUE
LongTerm: The starting tenor (inclusive) for Libor-OIS Spread Instruments	longterm	3Y	OIS SWAP	5Y	-0.271%			TRUE
LongTerm.GenerateMethod: The compounding method for Libor OIS Spread instruments. Can be "DailyAveraging" or remove this key and value for no compounding.	longterm.generatemethod	dailyaveraging	OIS SWAP	6Y	-0.176%			TRUE
Epsilon: The solver precision to be used when calibrating the	Epsilon	1.00E-10	OIS SWAP	7Y	-0.066%			TRUE
MaxLoop: The maximum number of iterations to allow when calibrating the curve	MaxLoop	1,000	OIS SWAP	8Y	0.055%			TRUE
SmoothShortEnd: This flag controls if instruments before the Central Bank Swaps such as O/N and T/N rates should be included in the calibration routine; True = Include instruments such as O/N and T/N before Central Bank Swaps and false will exclude such instruments.	SmoothShortEnd	TRUE	OIS SWAP	9Y	0.175%			TRUE
			OIS SWAP	10Y	0.284%			TRUE

Libor-OIS Basis Swaps

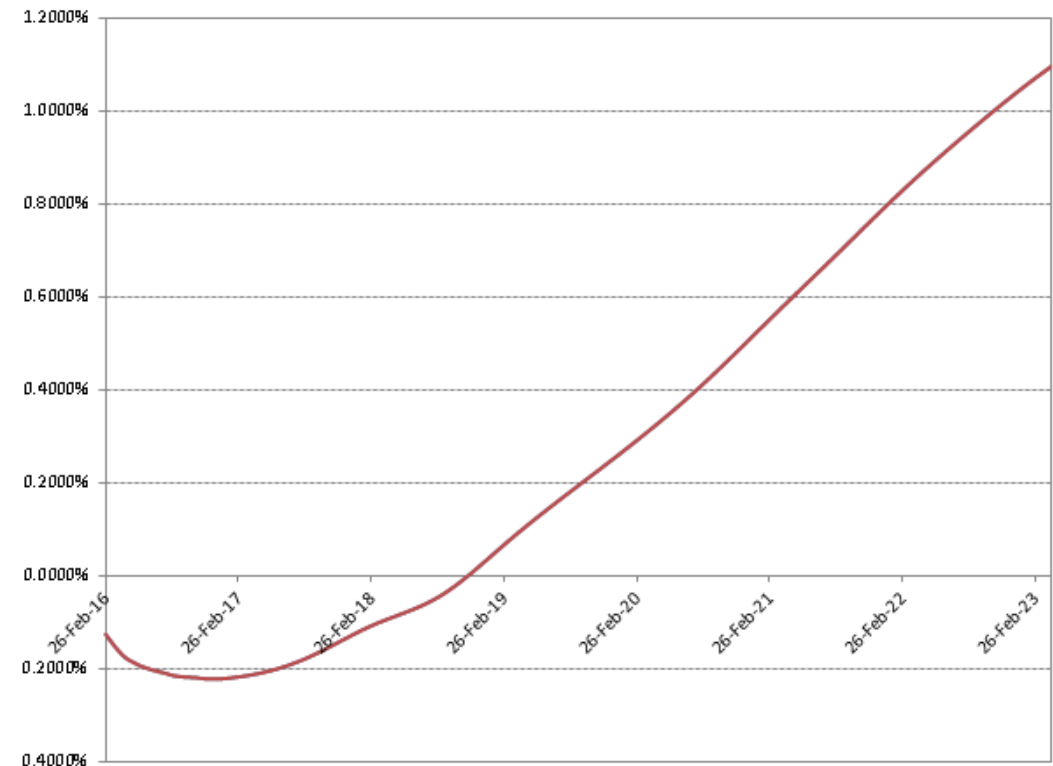
Calendar: The fixed leg calendar	Libor-OIS Tenor Basis Conventions		Libor-OIS Spreads							
Daycount: The fixed leg daycount						Calendar	TGT	Term	Rate	
Frequency: The fixed leg frequency						Daycount	ACT/360	LIBOR-OIS SPREAD	2Y	0.111%
SlidingRule: The fixed leg BusinessDayAdjustment						Frequency	Quarterly	LIBOR-OIS SPREAD	3Y	0.118%
Important Note: The LIBOR-OIS Spreads must be input in order and the Tenors must match those listed in the OIS Swap table	SlidingRule	MOD_FOLLOWING	LIBOR-OIS SPREAD	4Y	0.124%					
			LIBOR-OIS SPREAD	5Y	0.126%					
			LIBOR-OIS SPREAD	6Y	0.126%					
			LIBOR-OIS SPREAD	7Y	0.125%					
			LIBOR-OIS SPREAD	8Y	0.122%					
			LIBOR-OIS SPREAD	9Y	0.119%					
			LIBOR-OIS SPREAD	10Y	0.118%					
			LIBOR-OIS SPREAD	11Y	0.116%					
			LIBOR-OIS SPREAD	12Y	0.115%					
			LIBOR-OIS SPREAD	15Y	0.108%					
		LIBOR-OIS SPREAD	20Y	0.096%						
		LIBOR-OIS SPREAD	25Y	0.086%						
		LIBOR-OIS SPREAD	30Y	0.077%						
		LIBOR-OIS SPREAD	40Y	0.064%						
		LIBOR-OIS SPREAD	50Y	0.056%						

Calendar:		Libor Swap Conventions		Libor Swap Rates		
The fixed leg calendar		calendar	TGT		Term	Rate
Daycount:		daycountfix	30/360	EUR3ML SWAP	2Y	-0.293%
The fixed leg daycount		frequencyfix	Annual	EUR3ML SWAP	3Y	-0.271%
Frequency:		slidingrule	MOD_FOLLOWING	EUR3ML SWAP	4Y	-0.222%
SlidingRule:				EUR3ML SWAP	5Y	-0.148%
The fixed leg frequency				EUR3ML SWAP	6Y	-0.051%
BusinessDayAdjustment				EUR3ML SWAP	7Y	0.060%
				EUR3ML SWAP	8Y	0.180%
				EUR3ML SWAP	9Y	0.298%
				EUR3ML SWAP	10Y	0.408%
				EUR3ML SWAP	11Y	0.510%
				EUR3ML SWAP	12Y	0.601%
				EUR3ML SWAP	15Y	0.807%
				EUR3ML SWAP	20Y	0.981%
				EUR3ML SWAP	25Y	1.037%
				EUR3ML SWAP	30Y	1.053%
				EUR3ML SWAP	40Y	1.053%
				EUR3ML SWAP	50Y	1.003%

Important Note:
The LIBOR Swaps must be input in order and the tenors must match Libor-OIS spreads in the table to left

Calibration Instruments

- Libor Rates
- FRAs
- Futures
- Swaps



Curve Features

Standard swap curves are using the above instruments. Curves are built using global optimization techniques having an analytical risk Jacobian. Furthermore Tension features are available

```
meCurveCalibrateSwap("CurveCollection","StaticDataTable","CurveIndex","CurveBuildProperties","MoneyMarketConventions","LiborConventions","LiborFixingTable","SwapConventions","SwapMarketData",
FRAConventions,"FRAMarketData3M","FRAMarketData6M","FuturesConventions","FuturesRates","ConvexityAdjustmentConventions","ConvexityAdjustmentMarketData")
```

meCurveCalibrateSwap

Parameter Specifications	
meCurveCalibrateSwap	EURYC STD Curve has been set (0.01000 sec)
Curve Loaded	TRUE

Curve Names	
CurveCollection	EURYC
StaticDataTable	STD
IndexName(s)	STD:EUR3ML

CurveCollection: Name of the Curve Group, usually grouped by Currency
StaticDataTable: (Case Sensitive!) A hard-coded reference to a property file table used to lookup default curve paremeters if not specified
IndexName: A user defined name for the output curve index e.g. EUROIS, EUR1ML, EUR3ML, EUR6ML, EUR12ML ... Mutliple names and aliases allowed

Curve Build Properties

AsOfDate: The curve build Date	Curve Build Properties																
Currency: The curve currency	<table> <tr> <td>AsOfDate</td><td>31-May-16</td></tr> <tr> <td>Currency</td><td>EUR</td></tr> <tr> <td>interpolation</td><td>fn_SplineInterpolation</td></tr> <tr> <td>yieldgen.interpolation</td><td>fn_SplineInterpolation</td></tr> <tr> <td>isfrause</td><td>FALSE</td></tr> <tr> <td>isfutureuse</td><td>TRUE</td></tr> <tr> <td>Frequency</td><td>Quarterly</td></tr> <tr> <td>dfcurvename</td><td>OIS</td></tr> </table>	AsOfDate	31-May-16	Currency	EUR	interpolation	fn_SplineInterpolation	yieldgen.interpolation	fn_SplineInterpolation	isfrause	FALSE	isfutureuse	TRUE	Frequency	Quarterly	dfcurvename	OIS
AsOfDate	31-May-16																
Currency	EUR																
interpolation	fn_SplineInterpolation																
yieldgen.interpolation	fn_SplineInterpolation																
isfrause	FALSE																
isfutureuse	TRUE																
Frequency	Quarterly																
dfcurvename	OIS																
Interpolation: The interpolation method used for getting the OIS discount factors																	
YieldGen.Interpolation: The interpolation method used for generating the target curve i.e. interpolation method to use when calibrating this curve																	
isFraUse: Should FRA instruments be included in the calibration routine; TRUE = Include FRAs, FALSE = Exclude FRAs																	
isFutureUse: Should Futures instruments be included in the calibration routine; TRUE = Include Futures, FALSE = Exclude Futures																	
Frequency: The curve frequency, this controls which Libor Fixing should be associated with this curve.																	
dfCurveName: Here we specify which discount factor curve should be used. Can be "OIS" or if left-blank Libor discounting will be assumed. (This field must not be used on the OIS Curve)																	

Cash Instruments

	Money Market O/N and T/N Conventions		Libor Fixing Conventions		Libor Fixing Data	
Calendar: The fixed leg calendar	Calendar	TGT	Calendar	TGT	Term	Rate
Daycount: The fixed leg daycount	DayCount	ACT/360	Daycount	ACT/360	1M	-0.34900%
SlidingRule: The fixed leg BusinessDayAdjustment	Slidingrule	FOLLOWING	Frequency	Simple	3M	-0.26100%
			Resetlag	2	6M	-0.15300%
			SlidingRule	MOD_FOLLOWING	12M	-0.01500%

* These fixings are used in the tenor basis curves see '[SpotRateTerm](#)'

Futures Instruments

Calendar:

[The fixed leg calendar](#)

Daycount:

[The fixed leg daycount](#)

ResetLag:

[The float fixing lag](#)

SlidingRule:

[The fixed leg BusinessDayAdjustment](#)

MeanReversion:

The mean reversion parameter for the convexity adjustment model. Use 0% for Ho-Lee and any other value for Hull-White 1F convexity adjustments

ApplyTension:

Insert sythetic points between sucessessive futures to simulate tension

TensionGap:

The frequency in days for synthetic tension points

Futures Conventions

calendar	TGT
daycount	Act/360
resetlag	2
slidingrule	MOD_FOLLOWING
MeanReversion	3.00%
ApplyTension	TRUE
TensionGap	2

Futures Data

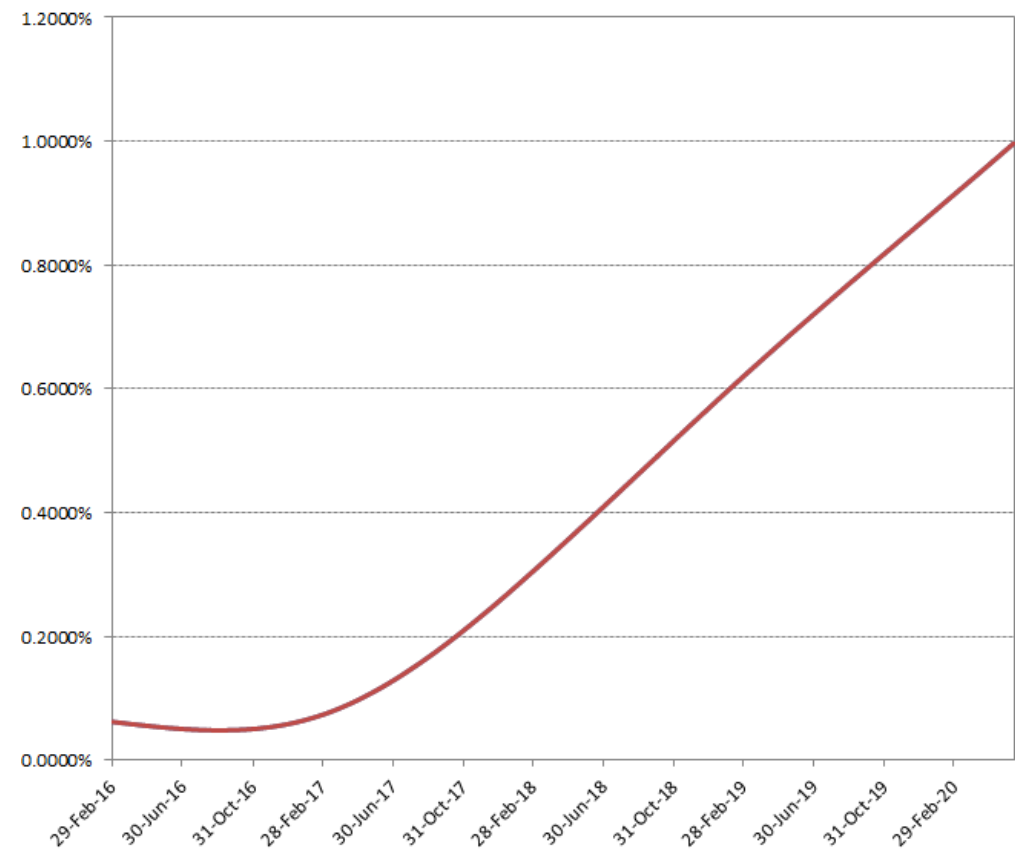
	Term / Ticker	StartDate	EndDate	Rate	Volatility
4	ERU6 Comdty	21-Sep-16	21-Dec-16	100.2800	0.514%
5	ERV6 Comdty	19-Oct-16	18-Jan-17	100.2850	0.514%
6	ERX6 Comdty	16-Nov-16	15-Feb-17	100.2900	0.514%
7	ERZ6 Comdty	21-Dec-16	15-Mar-17	100.2950	0.514%
8	ERH7 Comdty	15-Mar-17	21-Jun-17	100.3000	0.514%
9	ERM7 Comdty	21-Jun-17	20-Sep-17	100.3050	0.514%
10	ERU7 Comdty	20-Sep-17	20-Dec-17	100.3050	0.514%
11	ERZ7 Comdty	20-Dec-17	21-Mar-18	100.3000	0.514%
12	ERH8 Comdty	21-Mar-18	20-Jun-18	100.2850	0.514%

Swap Instruments

Swap Conventions			Swap Data		
Calendar: The fixed leg calendar	Calendar	TGT	Term	Rate	Use
Daycount: The fixed leg daycount	DayCount	30/360	3Y	-0.26500%	TRUE
DaycountFloat: The float leg daycount	DayCountFloat	ACT/360	4Y	-0.20500%	TRUE
Frequency: The fixed leg frequency	Frequency	Annual	5Y	-0.12475%	TRUE
FrequencyFloat: The float leg frequency	FrequencyFloat	Quarterly	6Y	-0.02475%	TRUE
Interpolation: The interpolation method used for getting the OIS discount factors	Interpolation	fn_SplineInterpolation	7Y	0.09025%	TRUE
IsEomRoll: Use the End of Month Roll Convention for swaps traded on End of Month	IsEomRoll	TRUE	8Y	0.21275%	TRUE
EomDay: Day to be considered month end	EomDay	31	9Y	0.33225%	TRUE
ResetLag: The float fixing lag	Resetlag	2	10Y	0.44425%	TRUE
SlidingRule: The fixed leg BusinessDayAdjustment	SlidingRule	MOD_FOLLOWING	11Y	0.54333%	TRUE
IsNewtonRaphson: Use Newton Raphson solver for calibration. Currently the only valid	IsNewtonraphson	TRUE	12Y	0.63458%	TRUE
IsSimultaneousEq: Solve the Newton Raphson problem as a Simultaneous Equation. Currently the	issimultaneouseq	TRUE	15Y	0.84325%	TRUE
Epsilon: The solver precision to be used when	Epsilon	1.00E-09	20Y	1.01675%	TRUE
MaxLoop: The maximum number of iterations to allow when calibrating the curve	MaxLoop	1,000	25Y	1.06967%	TRUE
			30Y	1.08675%	TRUE

Calibration Instruments

- Tenor Basis Swaps
- or Xccy Basis Swaps



Curve Features

Both Tenor & Xccy Basis Curves are built using bootstrapping / solver techniques

```
meCurveCalibrateBasis("CurveCollection","StaticDataTable","CurveIndex","CurveBuildProperties","BasisSwapConventions","BasisSwapMarketData",
FXForwardConventions,"FXForwardMarketData","SpotFXMarketData")
```

meCurveCalibrateBasis

Parameter Specifications	meCurveCalibrateBasis	EURYC 3M6MBasis Curve has been set (0.01000 sec)
CurveCollection: Name of the Curve Group, usually grouped by Currency	Curve Loaded	TRUE
StaticDataTable: (Case Sensitive!) A hard-coded reference to a property file table used to lookup default curve paremeters if not specified	Curve Names	
IndexName: A user defined name for the output curve index e.g. EUROIS, EUR1ML, EUR3ML, EUR6ML, EUR12ML ... Mutliple names and aliases allowed	CurveCollection	EURYC
	StaticDataTable	3M6MBasis
	IndexName(s)	3M6MBasis:EUR6ML

Curve Build Properties

Curve Build Properties	Curve Build Properties	Curve Build Properties
AsOfDate: The curve build Date	AsOfDate	31-May-16
Currency: The curve currency	Currency	EUR
Basis.Interpolation The interpolation method used for generating the target curve i.e. interpolation method to use when calibrating this curve	basis.interpolation	fn_SplineInterpolation
isFwdBasis: Allow forward starting tenor basis; No = Don't allow forward starting Tenor Basis Spreads, True will allow the extra columns for forward starting instruments (highlighted in yellow) in the Tenor Basis Spreads table to be recognized	isFwdBasis	TRUE
dfCurveName: Here we specify which discount factor curve should be used. Can be "OIS" or if left-blank Libor discounting will be assumed. (This field must not be used on the OIS Curve)	dfCurveName	OIS

Tenor Basis Spread Instruments I

Tenor Basis Swaps

MarketName:		* Don't mix dates and tenors: Supply StartDate and EndDate or StartTenor and EndTenor				
StaticDataTable name, a hard-coded reference to a property file table used to lookup default curve parameters if not specified						
IsLeg1Spread:						
Should the Tenor Basis spread be added to Leg 1 of the Tenor Basis Swap? True = Add spread to Leg1, False = Add spread to Leg2						
Leg1Cashlet.Calendar:						
Leg1 Accrual Calendar						
Leg1Cashlet.Daycount:						
Leg1 Accrual Daycount						
Leg1Cashlet.Frequency:						
Leg1 Accrual Frequency						
Leg1Cashlet.SlidingRule:						
Leg1 Accrual BusinessDayAdjustment or SlidingRule						
Leg1Cashlet.SpotLag:						
Leg1 Accrual Spot Lag (days)						
Leg1Discount:						
Leg1 Discount Curve e.g. EUROIS						
Leg1Forecast:						
Leg1 Forecast Curve e.g. EUR1ML, EUR3ML, EUR6ML, EUR12ML						
Leg1Index.Accessary:						
Leg1 Forecast Curve or Index Frequency						
Leg1Index.Daycount:						
Leg1 Payment Daycount						
Leg1Index.Fixing.Calendar:						
Leg1 Fixing Calendar						
Leg1Index.Frequency:						
Leg1 Yield Compound Frequency; Simple, Continuous, Annual, Semi-Annual, Quarterly etc						
Leg1Index.Payment.Calendar:						
Leg1 Payment Calendar						
Leg1Index.ResetLag:						
Leg1 Fixing Lag (days)						
Leg1Index.SlidingRule:						
Leg1 Payment BusinessDayAdjustment or SlidingRule						

Tenor Basis Conventions (3 X 6)

MarketName	EUR3M6MBASIS
IsLeg1Spread	TRUE
Leg1cashlet.calendar	TGT
Leg1cashlet.daycount	Act/360
Leg1cashlet.frequency	Quarterly
Leg1cashlet.slidingrule	MOD_FOLLOWING
Leg1cashlet.spotlag	2
Leg1discount	OIS
Leg1forecast	EUR3ML
Leg1index.accessary	3M
Leg1index.daycount	Act/360
Leg1index.fixing.calendar	TGT
Leg1index.frequency	Simple
Leg1index.payment.calendar	TGT
Leg1index.resetlag	2
Leg1index.slidingrule	MOD_FOLLOWING

Tenor Basis Spreads (3 X 6)

Term	Rate	isDate	StartDate or Tenor	EndDate or Tenor
6M	0.13000%	TRUE	02-Jun-16	02-Dec-16
6M	0.13000%	TRUE	02-Dec-16	02-Jun-17
6M	0.13000%	TRUE	02-Jun-17	04-Dec-17
6M	0.13000%	TRUE	04-Dec-17	04-Jun-18
3Y	0.13000%	FALSE	0y	3Y
4Y	0.13500%	FALSE	0y	4Y
5Y	0.13900%	FALSE	0y	5Y
6Y	0.13900%	FALSE	0y	6Y
7Y	0.13700%	FALSE	0y	7Y
8Y	0.13250%	FALSE	0y	8Y
9Y	0.12600%	FALSE	0y	9Y
10Y	0.11900%	FALSE	0y	10Y
11Y	0.11250%	FALSE	0y	11Y
12Y	0.10650%	FALSE	0y	12Y
15Y	0.09100%	FALSE	0y	15Y

Tenor Basis Spread Instruments II

Leg2Cashlet.Calendar: Leg2 Accrual Calendar	Leg2cashlet.calendar	TGT	20Y	0.07100%	FALSE	0y	20Y
Leg2Cashlet.Daycount: Leg2 Accrual Daycount	Leg2cashlet.daycount	Act/360	25Y	0.05825%	FALSE	0y	25Y
Leg2Cashlet.Frequency: Leg2 Accrual Frequency	Leg2cashlet.frequency	Semi-Annual	30Y	0.04900%	FALSE	0y	30Y
Leg2Cashlet.SlidingRule: Leg2 Accrual BusinessDayAdjustment or SlidingRule	Leg2cashlet.slidingrule	MOD_FOLLOWING					
Leg2Cashlet.SpotLag: Leg2 Accrual Spot Lag (days)	Leg2cashlet.spotlag	2					
Leg2Discount: Leg2 Discount Curve e.g. EUROIS	Leg2discount	OIS					
Leg2Forecast: Leg2 Forecast Curve e.g. EUR1ML, EUR3ML, EUR6ML, EUR12ML	Leg2forecast	STD			← Solver: Initial Guess		
Leg2Index.Accessary: Leg2 Forecast Curve or Index Frequency	Leg2index.accessary	6M					
Leg2Index.Daycount: Leg2 Payment Daycount	Leg2index.daycount	Act/360					
Leg2Index.Fixing.Calendar: Leg2 Fixing Calendar	Leg2index.fixing.calendar	TGT					
Leg2Index.Frequency: Leg2 Yield Compound Frequency; Simple, Continuous, Annual, Semi-Annual, Quarterly etc	Leg2index.frequency	Simple					
Leg2Index.Payment.Calendar: Leg2 Payment Calendar	Leg2index.payment.calendar	TGT					
Leg2Index.ResetLag: Leg2 Fixing Lag (days)	Leg2index.resetlag	2					
Leg2Index.SlidingRule: Leg2 Payment BusinessDayAdjustment or SlidingRule	Leg2index.slidingrule	MOD_FOLLOWING					

Tenor Basis Spread Instruments III

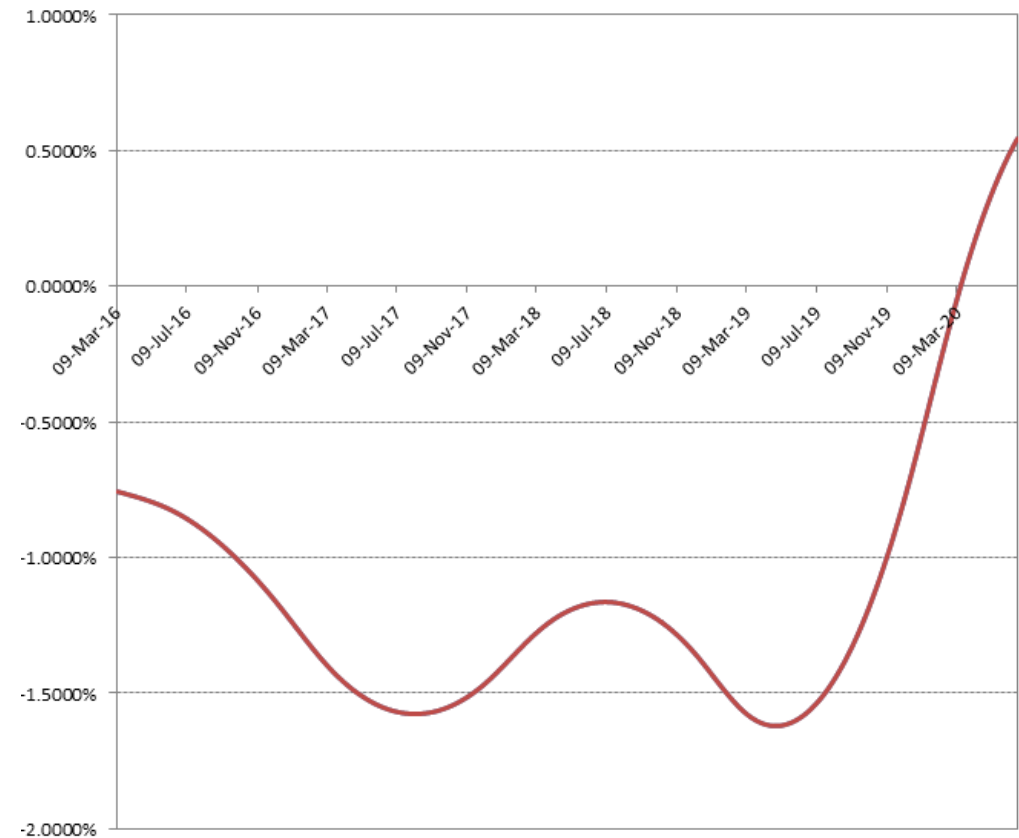
Target: What do we want to solve for	Target	Leg2Forecast	<--- Solver: Target
IsSimultaneousEq: Calibrate and optimize using a Simultaneous Equation method. Currently the only valid option is "TRUE".	issimultaneouseq	TRUE	
FwdInterpolation: Interpolation method for interpolating Libor forecast rates	fwdinterpolation	fn_SplineInterpolation	
IsFwdInter: Imply Libor Forecast rates by interpolating forwards; True = Imply forecast rates from Forwards, False = Imply forecast rates from Pseudo-Discount Factors	isfwdinter	TRUE	
IsSameGridIndex: Are the index fixing dates the same as the cashflow accrual dates i.e are the date grids the same? Recommended setting is "TRUE", sometimes when set to false curves will fail to solve and converge.	issamegridindex	TRUE	
SpotRateTerm: Tenor Basis curve fixings must be input in the Swap or STD curve Libor Fixing table. This field specifies which Libor fixing should be used when building the Tenor Basis Curve. Select 3M if building a 3M curve and 6M if building a 6M curve etc.	spotrateterm	6M	
isYieldSpreadCalc: Is the calibration target a spread on a base curve. This field should typically be set to "FALSE"	isyieldspreadcalc	FALSE	
Epsilon: The solver precision to be used when calibrating the curve	Epsilon	1.00E-09	
MaxLoop: The maximum number of iterations to allow when calibrating the curve	MaxLoop	1,000	

Build Method

- Bootstrapping
- Assumes constant FX Forwards or **FX Forward Invariance**

Calibration Instruments

- Xccy Basis Swaps
- FX Forwards



Curve Features

CSA curves are bootstrapped from Xccy swaps and assume FX Forward invariance, see next slide.

- We assume that FX Forwards are constant
- FX Forwards are fixed assuming a USD CSA

FX Forward Invariance - Example

$$FwdFX(t, T)^{EUR/GBP} = S \frac{P(t, T)^{EUR_USDCSA}}{P(t, T)^{GBP_USDCSA}} = S \frac{P(t, T)^{EUR_GBPCSA}}{P(t, T)^{GBP_GBPCSA}}$$

where t denotes the valuation date, S the FX spot rate, $P(t, T)$ the discount factor at time t for tenor T with $0 \leq t < T$

- From which Discounted CSA Curves can then be implied

Discount CSA Curve - General Formula

$$P(t, T)^{Ccy1_Ccy2CSA} = \left(\frac{P(t, T)^{Ccy1_USDCSA}}{P(t, T)^{Ccy2_USDCSA}} \right) \times \underbrace{P(t, T)^{Ccy2_Ccy2CSA}}_{\text{Ccy2 OIS Discount Factor}}$$

- Forwards can be retrieved using `meCurveForwardRates`
- Discount Factors can be retrieved using `meCurveDisplay` or `meCurveDiscountFactors`

Curve ID	EURYC
Curve Name	EUR1ML
Description	1ML Projection Curve

Curve ID	EURYC
Curve Name	EUR3ML
Description	3ML Projection Curve

Curve ID	EURYC
Curve Name	EUR6ML
Description	6ML Projection Curve

Curve ID	EURYC
Curve Name	EUR12ML
Description	12ML Projection Curve

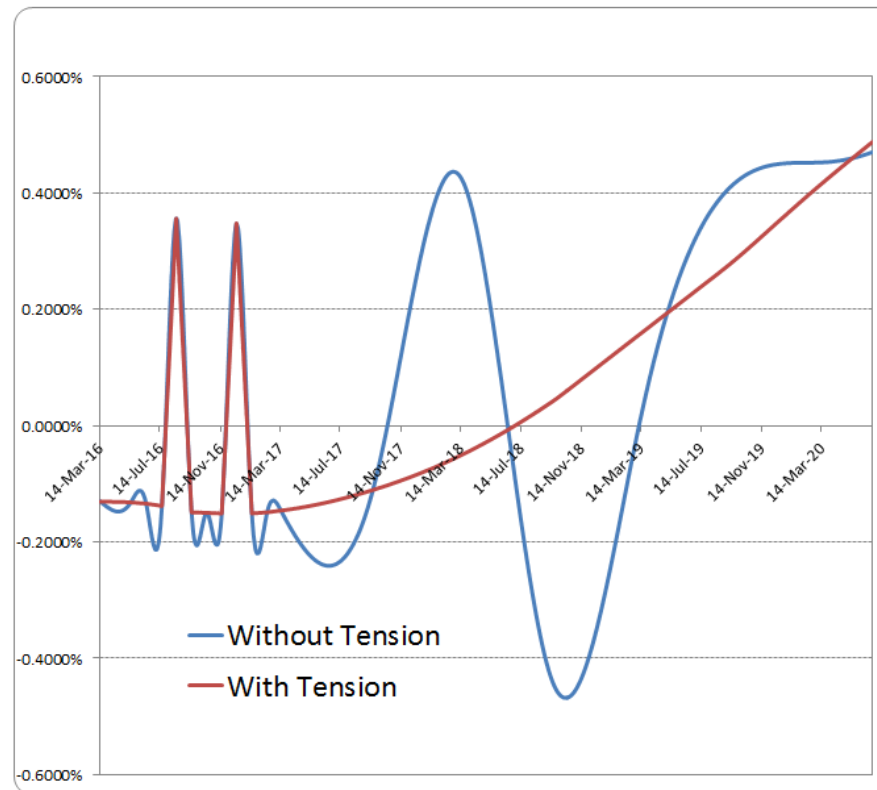
Curve ID	EURYC
Curve Name	EURDF_NOCSA
Description	Discount Curve No CSA

Term	DiscFact	Term	DiscFact	Term	DiscFact	Term	DiscFact	Term	DiscFact
0.00	1.000000	0.00	1.000000	0.00	1.000000	0.00	1.000000	0.00	1.000000
0.01	1.000045	0.01	1.000030	0.01	1.000010	0.01	1.000002	0.01	1.000020
0.10	1.000390	0.26	1.000732	0.01	1.000014	1.01	1.000227	0.01	1.000031
0.18	1.000748	0.51	1.001452	0.18	1.000229	2.01	1.000561	0.01	1.000041
0.26	1.001073	0.76	1.002198	0.19	1.000235	3.01	0.999824	0.02	1.000051
0.34	1.001412	1.01	1.002990	0.19	1.000241	4.02	0.997055	0.02	1.000061
0.43	1.001777	1.26	1.003859	0.20	1.000247	5.01	0.991868	0.03	1.000092
0.51	1.002123	1.51	1.004757	0.20	1.000253	6.01	0.984171	0.03	1.000102
0.60	1.002473	1.76	1.005634	0.21	1.000258	7.01	0.973544	0.03	1.000113
0.68	1.002840	2.01	1.006449	0.21	1.000264	8.01	0.960571	0.04	1.000123
0.76	1.003201	2.26	1.007196	0.22	1.000269	9.01	0.946044	0.04	1.000133
0.85	1.003606	2.51	1.007842	0.22	1.000274	10.01	0.930846	0.05	1.000164
0.93	1.003970	2.76	1.008378	0.23	1.000280	11.01	0.914870	0.05	1.000174
1.01	1.004328	3.01	1.008800	0.23	1.000285	12.01	0.898666	0.05	1.000184
1.10	1.004785	3.26	1.009115	0.24	1.000290	13.01	0.882585	0.05	1.000195
1.18	1.005145	3.52	1.009302	0.25	1.000295	14.01	0.866796	0.06	1.000205
1.26	1.005549	3.77	1.009338	0.25	1.000300	15.01	0.851590	0.07	1.000236
1.34	1.005958	4.02	1.009220	0.26	1.000305	16.01	0.837089	0.07	1.000246
1.43	1.006382	4.27	1.008935	0.26	1.000310	17.01	0.823492	0.07	1.000256
1.51	1.006808	4.51	1.008479	0.27	1.000315	18.01	0.810540	0.07	1.000266
1.60	1.007246	4.76	1.007848	0.27	1.000320	19.01	0.798253	0.08	1.000277
1.68	1.007641	5.01	1.007037	0.28	1.000325	20.01	0.786558	0.08	1.000307
1.76	1.008047	5.26	1.006041	0.28	1.000330	21.01	0.775425	0.09	1.000318
1.85	1.008473	5.51	1.004835	0.29	1.000335	22.01	0.764878	0.09	1.000328
1.93	1.008865	5.76	1.003435	0.30	1.000339	23.01	0.754799	0.09	1.000338
2.01	1.009224	6.01	1.001833	0.30	1.000344	24.01	0.745093	0.11	1.000389
2.10	1.009636	6.26	0.999968	0.31	1.000349	25.01	0.735804	0.11	1.000400
2.18	1.009977	6.51	0.997886	0.31	1.000354	26.01	0.726826	0.11	1.000410

Current interpolation methods for curve construction:

- Piecewise Constant [fn_stepInterpolation](#)
- Linear [fn_linearInterpolation](#)
- Monotone Convex [fn_monotoneconvexInterpolation](#)
- Spline [fn_splineInterpolation](#)
- Constrained Spline [fn_constrainedsplineInterpolation](#)

- Curve tension can be applied to swap curves
- Can be added to FRAs and / or Futures Instruments
- Used to manage curve over-oscillation & market data shocks



- All curves use optimized using Newton-Raphson like solvers
- First-order risk matrices or **Jacobians** are a by-product

Newton-Raphson Optimization

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where $f'(x_n)$ is our first order sensitivity

- Curves are being extended to store and use this data
- The Jacobian is useful for analytical risk and speedy curve re-builds

Risk Jacobian

Curves are globally optimized having analytical Jacobian risk metrics, $\mathcal{J}$

$$\underline{L}_{n+1} = \underline{L}_n - \mathcal{J}F(\underline{L}_n)$$

where $\underline{L}$ is our Libor vector, $F(\underline{L})$ the calibration pricing kernel and $\mathcal{J} = F'(\underline{L}_n)^{-1}$ is the Jacobian or sensitivity matrix

mirSetUpOISCurve (CurveCollection, StaticDataTable, CurveBuildProperties, OISConventions, OISMktData, IndexName, OISFixingTable, OISBasisMktData, SwapConventions, SwapMktData, OISBasisConventions)

mirSetUpOISCurve

mirSetUpOISCurve	OIS OIS Curve has been set (0.04700 sec)
Curve Names	
CurveCollection	EURYC
StaticDataTable	OIS
IndexName(s)	OIS:EURDF

Curve Build Properties

Curve Build Properties		Interpolation Options
AsOfDate	19-Feb-16	fn_LinearInterpolation
Currency	EUR	fn_StepInterpolation
yieldgen.interpolation	fn_constrainedsplineInterpolation	fn_MonotoneConvexInterpolation
		fn_SplineInterpolation
		fn_ConstrainedSplineInterpolation

Overnight Indexed Swaps

OIS Conventions		OIS Par Rates					OIS Fixing Table	
calendar	TGT	Term	Rate	StartDate	EndDate	Use	Date	Rate
daycount	ACT/360	ON	-0.244%			TRUE		
frequency	Annual	EUSF1A	-0.369%	10-Mar-16	21-Apr-16	TRUE		
resetlag	2d	EUSF2A	-0.392%	21-Apr-16	02-Jun-16	TRUE		
slidingrule	MOD_FOLLOWING	EUSF3A	-0.445%	02-Jun-16	21-Jul-16	TRUE		
iseomroll	TRUE	EUSF4A	-0.453%	21-Jul-16	08-Sep-16	TRUE		
eomday	31	EUSF5A	-0.476%	08-Sep-16	20-Oct-16	TRUE		
generatemethod	dailycompounding	12M	-0.433%			TRUE		
firstrate	ShortMarket	18M	-0.448000%			TRUE		
shorttermconvention	arithmeticaverage	2Y	-0.448%			TRUE		
longtermconvention	LOBASIS	3Y	-0.409%			TRUE		
longterm	3Y	4Y	-0.337%			TRUE		
longterm.generatemethod	dailyaveraging	5Y	-0.249%			TRUE		
		6Y	-0.140%			TRUE		
		7Y	-0.024%			TRUE		
		8Y	0.093%			TRUE		
		9Y	0.205%			TRUE		
		10Y	0.305%			TRUE		

Libor-OIS Basis Swaps

Libor-OIS Tenor Basis Conventions		Libor-OIS Spreads		Libor Swap Conventions		Libor Swap Rates	
Calendar	TGT	Term	Rate	calendar	TGT	Term	Rate
Daycount	ACT/360	2Y	0.155%	daycountfix	30/360	2Y	-0.295%
Frequency	Quarterly	3Y	0.159%	frequencyfix	Annual	3Y	-0.254%
SlidingRule	MOD_FOLLOWING	4Y	0.162%	slidingrule	MOD_FOLLOWING	4Y	-0.178%
		5Y	0.165%			5Y	-0.085%
		6Y	0.163%			6Y	0.023%
		7Y	0.162%			7Y	0.140%
		8Y	0.161%			8Y	0.258%
		9Y	0.160%			9Y	0.370%
		10Y	0.159%			10Y	0.471%

mirSetUpOISCurve (CurveCollection, StaticDataTable, CurveBuildProperties, MoneyMktConventions, LiborConventions, LiborMktData, SwapConventions, SwapMktData, FRAConventions, FRAMktData3M, FRAMktData6M, FuturesConventions, FuturesMktData, [QuantoConventions], [QuantoMktData], IndexName, [TradeDiscountCurve])

mirSetUpSwapCurve

mirSetUpSwapCurve	STD Swap Curve has been set (0.01600 sec)
Curve Loaded	TRUE
Curve Names	
CurveCollection	EURYC
StaticDataTable	STD
IndexName(s)	STD:EUR6ML

Curve Build Properties

Curve Build Properties		Interpolation Options
AsOfDate	19-Feb-16	fn_LinearInterpolation
Currency	EUR	fn_StepInterpolation
future.interpolation	fn_MonotoneConvexInterpolation	fn_MonotoneConvexInterpolation
interpolation	fn_MonotoneConvexInterpolation	fn_SplineInterpolation
yieldgen.interpolation	fn_MonotoneConvexInterpolation	fn_ConstrainedSplineInterpolation
isfrause	TRUE	
isfutureuse	FALSE	
Frequency	Semi-Annual	
dfcurvename	OIS	

Cash Instruments

Money Market Conventions		Libor Conventions		Libor Market Data	
Calendar	TGT	Calendar	TGT	Term	Rate
DayCount	ACT/360	Daycount	ACT/360	6M	-0.13100%
Slidingrule	FOLLOWING	Frequency	Simple		
		Resetlag	2		
		SlidingRule	MOD_FOLLOWING		

FRA Instruments

FRA Conventions		FRA Market Data X3		FRA Market Data x6	
Calendar	TGT	Term	Rate	Term	Rate
DayCount	ACT/360			2X8	-0.13400%
ResetLag	2			3X9	-0.13600%
SlidingRule	MOD_FOLLOWING			4X10	-0.14200%
IsEomRoll	FALSE			5X11	-0.14800%
ApplyTension	TRUE			6X12	-0.15400%
TensionGap	2			7X13	-0.15700%
				8X14	-0.16000%
				9X15	-0.16300%
				10X16	-0.16300%
				11X17	-0.16200%
				12X18	-0.16200%

Vanilla Addin: Example Swap Curve III

Futures Instruments

Futures Conventions

calendar	NYB
daycount	Act/360
resetlag	2
slidingrule	MOD_FOLLOWING
MeanReversion	3.00%
ApplyTension	TRUE
TensionGap	2

Futures Data

Term / Ticker	StartDate	EndDate	Rate	Volatility
ED2 COMB Comdty	21-Sep-16	21-Dec-16	99.1700	0.900%
ED3 COMB Comdty	21-Dec-16	15-Mar-17	99.0850	0.900%
ED4 COMB Comdty	15-Mar-17	21-Jun-17	99.0250	0.900%
ED5 COMB Comdty	21-Jun-17	20-Sep-17	98.9600	0.900%
ED6 COMB Comdty	20-Sep-17	20-Dec-17	98.8850	0.900%
ED7 COMB Comdty	20-Dec-17	21-Mar-18	98.8100	0.900%
ED8 COMB Comdty	21-Mar-18	20-Jun-18	98.7500	0.900%
ED9 COMB Comdty	20-Jun-18	19-Sep-18	98.6800	0.900%
ED10 COMB Comdty	19-Sep-18	19-Dec-18	98.6150	0.900%
ED11 COMB Comdty	19-Dec-18	20-Mar-19	98.5400	0.900%
ED12 COMB Comdty	20-Mar-19	19-Jun-19	98.4850	0.900%

Swap Instruments

Swap Conventions

Calendar	LNB:NYB
DayCount	30/360
DayCountFix	30/360
DayCountFloat	ACT/360
Frequency	Semi-Annual
FrequencyFloat	Quarterly
Interpolation	fn_MonotoneConvexInterpolation
IsEomRoll	FALSE
IsNewtonraphson	TRUE
IsTimeInterpolation	TRUE
Resetlag	2
SlidingRule	MOD_FOLLOWING
issimultaneousseq	TRUE
optimizemethod	NL2SOL
Epsilon	1.00E-09
GradientEpsilon	1.00E-15
Delta	1.00E-10
MaxLoop	1,000

Swap Data

Term	Rate	Use
5Y	1.30050%	TRUE
7Y	1.50200%	TRUE
10Y	1.73450%	TRUE
11Y	1.79900%	TRUE
12Y	1.84950%	TRUE
15Y	1.97200%	TRUE
20Y	2.09750%	TRUE
25Y	2.15800%	TRUE
30Y	2.19800%	TRUE
40Y	2.21520%	TRUE

mirSetUpBasisCurve (CurveCollection, StaticDataTable, CurveBuildProperties, BasisSwapConventions, BasisSwapMarketData, IndexName, [ForwardFXConventions], [ForwardFXMarketData], [SpotFXMarketData])

mirSetUpBasisSwapCurve

mirSetUpBasisSwapCurve 3M6MBasis BasisSwap Curve has been set (0.01500 sec)
 Prior to building the Basis Curve must construct an OIS Curve using 'mirSetUpOISCurve' and a Swap Curve using 'mirSetUpSwapCurve'

Curve Names

CurveCollection	EURYC
StaticDataTable	3M6MBasis
IndexName(s)	3M6MBasis:EUR3ML

<- This is a reference to a property file table and should remain fixed.

Curve Build Properties

Curve Build Properties

AsOfDate	19-Feb-16
Currency	EUR
basis.interpolation.3m6mbasis	fn_MonotoneConvexInterpolation
basis.interpolation.1m3mbasis	fn_MonotoneConvexInterpolation
basis.interpolation.6m12mbasis	fn_MonotoneConvexInterpolation

Interpolation Options

fn_LinearInterpolation
fn_StepInterpolation
fn_MonotoneConvexInterpolation
fn_SplineInterpolation
fn_ConstrainedSplineInterpolation

Vanilla Addin: Example Tenor-Basis Curve II

Tenor Basis Conventions (3 X 6)			Tenor Basis Spreads (3 X 6)		
Known Leg	MarketName	EUR3M6MBASIS	Term	Rate	
	IsLeg1Spread	FALSE	1Y	0.11300%	
	Leg1cashlet.calendar	TGT	2Y	0.11925%	
	Leg1cashlet.daycount	Act/360	3Y	0.12425%	
	Leg1cashlet.frequency	Semi-Annual	4Y	0.12975%	
	Leg1cashlet.slidingrule	MOD_FOLLOWING	5Y	0.13325%	
	Leg1cashlet.spotlag	2	6Y	0.13075%	
	Leg1discount	OIS	7Y	0.12725%	
	Leg1forecast	EUR6ML	8Y	0.12275%	
	Leg1index.accessary	6M	9Y	0.11675%	
	Leg1index.daycount	Act/360	10Y	0.11025%	
	Leg1index.fixing.calendar	TGT	11Y	0.10425%	
	Leg1index.frequency	Simple	12Y	0.09925%	
	Leg1index.payment.calendar	TGT	15Y	0.08625%	
	Leg1index.resetlag	2	20Y	0.06875%	
	Leg1index.slidingrule	MOD_FOLLOWING	25Y	0.05825%	
Leg2Forecast Unknown : STD is the initial guess	Leg2cashlet.calendar	TGT	30Y	0.05125%	
	Leg2cashlet.daycount	Act/360			
	Leg2cashlet.frequency	Quarterly			
	Leg2cashlet.slidingrule	MOD_FOLLOWING			
	Leg2cashlet.spotlag	2			
	Leg2discount	OIS			
	Leg2forecast	STD			
	Leg2index.accessary	3M			
	Leg2index.daycount	Act/360			
	Leg2index.fixing.calendar	TGT			
	Leg2index.frequency	Simple			
	Leg2index.payment.calendar	TGT			
	Leg2index.resetlag	2			
	Leg2index.slidingrule	MOD_FOLLOWING			
	Solving for : Leg2Forecast Rate	Target	Leg2Forecast		
		issimultaneouseq	TRUE		
fwdinterpolation		fn_MonotoneConvexInterpolation			
isfwdinter		TRUE			
issamegridindex		TRUE			
spotrateterm		3M			
isyieldspreadcalc		FALSE			
Epsilon		1.00E-09			
GradientEpsilon		1.00E-10			
Delta		1.00E-10			
MaxLoop	1,000				

mirSetUpBasisCurve (CurveCollection, StaticDataTable, CurveBuildProperties, BasisSwapConventions, BasisSwapMarketData, IndexName, [ForwardFXConventions], [ForwardFXMarketData], [SpotFXMarketData])

mirSetUpBasisSwapCurve

mirSetUpBasisSwapCurve XccyBasis BasisSwap Curve has been set (0.01600 sec)
Prior to building the Basis Curve must construct an OIS Curve using 'mirSetUpOISCurve' and a Swap Curve using 'mirSetUpSwapCurve'

Curve Names

CurveCollection	EURYC
StaticDataTable	XccyBasis
IndexName(s)	XccyBasis:EURDF_USDCSA

<- This is a reference to a property file table and should remain fixed.

Curve Build Properties

Curve Build Properties

AsOfDate	19-Feb-16
Currency	EUR
basis.interpolation.xccybasis	fn_MonotoneConvexInterpolation

Interpolation Options

fn_LinearInterpolation
fn_StepInterpolation
fn_MonotoneConvexInterpolation
fn_SplineInterpolation
fn_ConstrainedSplineInterpolation

Vanilla Addin: Example Xccy-Basis Curve II

Xccy Basis Swap Conventions			Xccy Basis Spreads	
Leg1Discount Unknown : OIS is the initial guess	MarketName	XCCYBASIS	Term	Rate
	adjustvalue.interpolation	fn_MonotoneConvexInterpolation	1Y	0.11300%
	isleg1spread	TRUE	2Y	0.11925%
	leg1cashlet.calendar	LNB:NYB:TGT	3Y	0.12425%
	leg1cashlet.daycount	Act/360	4Y	0.12975%
	leg1cashlet.frequency	Quarterly	5Y	0.13325%
	leg1cashlet.slidingrule	MOD_FOLLOWING	6Y	0.13075%
	leg1cashlet.spotlag	2d	7Y	0.12725%
	leg1discount	EURYC:OIS	8Y	0.12275%
	leg1forecast	EURYC:EUR3ML	9Y	0.11675%
	leg1index.accessary	3M	10Y	0.11025%
	leg1index.daycount	Act/360	11Y	0.10425%
	leg1index.fixing.calendar	TGT	12Y	0.09925%
	leg1index.frequency	Simple	15Y	0.08625%
	leg1index.payment.calendar	LNB	20Y	0.06875%
	leg1index.resetlag	2	25Y	0.05825%
	leg1index.slidingrule	MOD_FOLLOWING	30Y	0.05125%
Known Leg	leg2cashlet.calendar	LNB:NYB:TGT		
	leg2cashlet.daycount	Act/360		
	leg2cashlet.frequency	Quarterly		
	leg2cashlet.slidingrule	MOD_FOLLOWING		
	leg2cashlet.spotlag	2		
	leg2discount	USDYC:OIS		
	leg2forecast	USDYC:USD3ML		
	leg2index.accessary	3M		
	leg2index.daycount	Act/360		
	leg2index.fixing.calendar	LNB		
	leg2index.frequency	Simple		
	leg2index.payment.calendar	LNB		
	leg2index.resetlag	2		
	leg2index.slidingrule	MOD_FOLLOWING		
Solving for : Leg1Discount Rate	target	Leg1Discount		
	issimultaneouseq	TRUE		
	isfwdinter	TRUE		
	fwdinterpolation	fn_MonotoneConvexInterpolation		
	isyieldspreadcalc	FALSE		
	spotrateterm	3M		
	Epsilon	1.00E-09		
	GradientEpsilon	1.00E-10		
	Delta	1.00E-10		
MaxLoop	1,000			

`mirSetUpFwdFXConstantCurve` (CurveCollection, StaticDataTable, CurveBuildProperties, FXForwardConventions, IndexName)

mirSetUpFwdFXConstantCurve

<code>mirSetUpFwdFXConstantCurve</code>	FwdFXConstGBP FwdFXConstant Curve has been set (0.00000 sec)
Curve Loaded	TRUE
Curve Names	
CurveCollection	EURYC
StaticDataTable	FwdFXConstGBP
IndexName(s)	FwdFXConstGBP:EURDF_GBPCSA

Curve Build Properties

Curve Build Properties	
AsOfDate	19-Feb-16
Currency	EUR

FX Forwards

FX Forward Conventions	
leg1discount	GBPYP:GBPDF
leg1forecast	GBPYP:GBPDF_USDCSA
leg2discount	EURYC:XccyBasis
leg2forecast	EURYC:EURDF_USDCSA
target	Leg2Discount

$$P(t, T)^{EUR_GBPCSA} = \left(\frac{P(t, T)^{EUR_USDCSA}}{P(t, T)^{GBP_USDCSA}} \right) \times P(t, T)^{GBP_OIS}$$

Build Order Native CSA Curves

- ① OIS Curve
- ② Standard Swap Curve(s)
- ③ Tenor Basis Curve(s)

OIS Discounting & CSA Curves

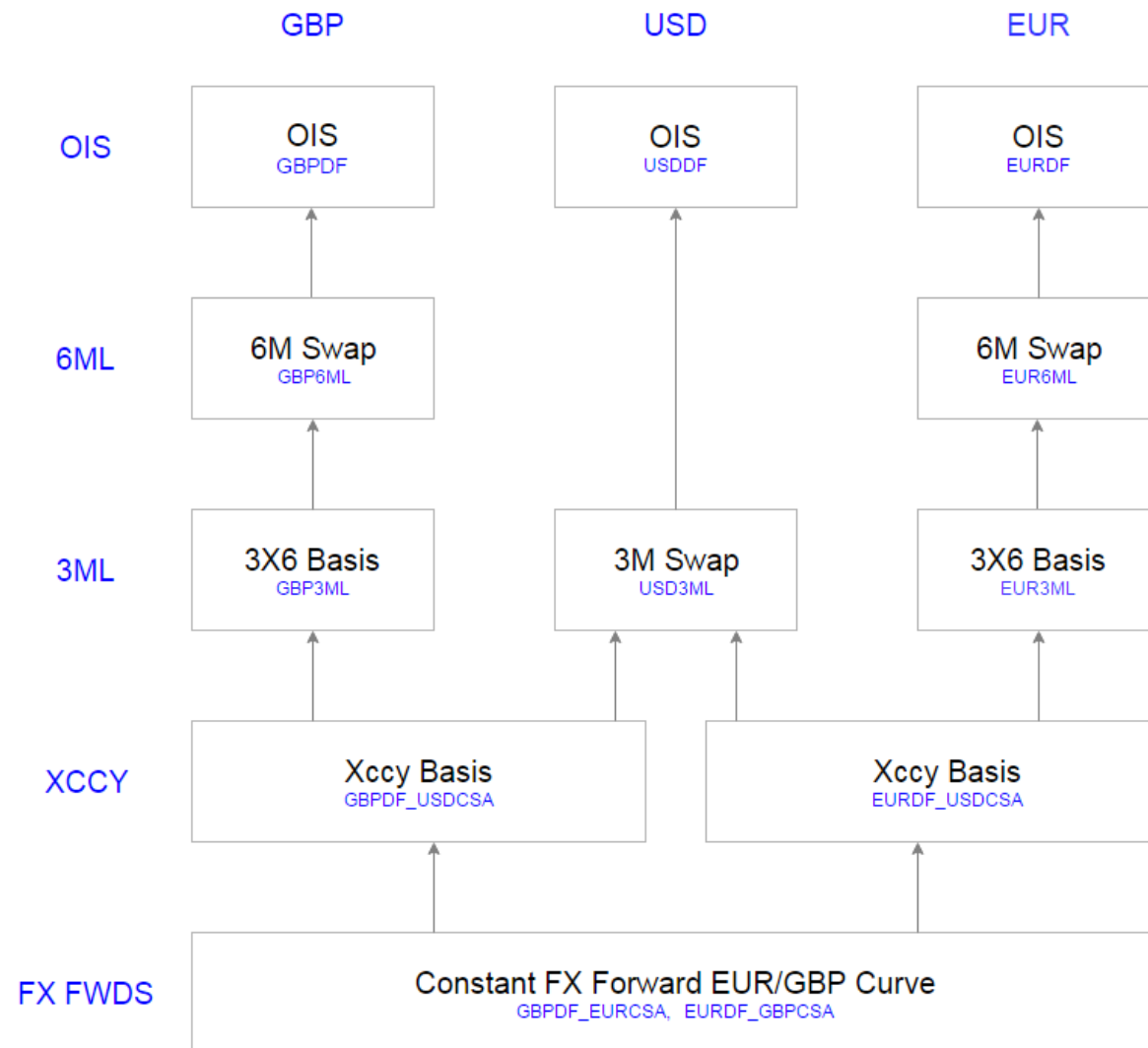
USD

	Instruments Used	USDYC
OIS	FED Funds vs 12m Fixed	OIS OIS Curve has been set (0.06200 sec)
STD	3m Float vs 6m Fixed	STD Swap Curve has been set (0.00000 sec)
3M6MBasis	Tenor Basis 3x6	3M6MBasis BasisSwap Curve has been set (0.01500 sec)
1M3MBasis	Tenor Basis 1x3	1M3MBasis BasisSwap Curve has been set (0.01600 sec)
3M12MBasis	Tenor Basis 3x12	3M12MBasis BasisSwap Curve has been set (0.00000 sec)

EUR

	Instruments Used	EURYC
OIS	EONIA vs 12m Fixed	OIS OIS Curve has been set (0.03100 sec)
STD	6m Float vs 12m Fixed	STD Swap Curve has been set (0.01500 sec)
3M6MBasis	Tenor Basis 3x6	3M6MBasis BasisSwap Curve has been set (0.00000 sec)
1M3MBasis	Tenor Basis 1x3	1M3MBasis BasisSwap Curve has been set (0.01600 sec)
3M12MBasis	Tenor Basis 6x12	3M12MBasis BasisSwap Curve has been set (0.00000 sec)

Non-Native CSA Curves have Additional Currency Dependencies



Swap Pricing, Native CSA

EUR Swaps

PV

-312,926

EUR Calibration Instruments

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	Float First	CSA	Par Rate	Present Value	PV01
1	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	1Y	1.00000%	6ML	0.00		NOCSA	-0.13969%	-11,440	100
2	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	2Y	1.00000%	6ML	0.00		NOCSA	-0.13300%	-22,791	201
3	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	3Y	1.00000%	6ML	0.00		NOCSA	-0.08700%	-32,910	303
4	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	4Y	1.00000%	6ML	0.00		NOCSA	-0.00800%	-40,682	404
5	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	5Y	1.00000%	6ML	0.00		NOCSA	0.09300%	-45,732	504
6	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	6Y	1.00000%	6ML	0.00		NOCSA	0.20200%	-48,247	605
7	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	7Y	1.00000%	6ML	0.00		NOCSA	0.32100%	-47,815	704
8	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	8Y	1.00000%	6ML	0.00		NOCSA	0.44400%	-44,630	803
9	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	9Y	1.00000%	6ML	0.00		NOCSA	0.56100%	-39,518	900
10	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	10Y	1.00000%	6ML	0.00		NOCSA	0.66700%	-33,156	996
11	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	11Y	1.00000%	6ML	0.00		NOCSA	0.76300%	-25,831	1,090
12	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	12Y	1.00000%	6ML	0.00		NOCSA	0.84700%	-18,094	1,183
13	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	13Y	1.00000%	6ML	0.00		NOCSA	0.91775%	-10,476	1,274
14	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	14Y	1.00000%	6ML	0.00		NOCSA	0.97836%	-2,951	1,364
15	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	15Y	1.00000%	6ML	0.00		NOCSA	1.03000%	4,355	1,452
16	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	16Y	1.00000%	6ML	0.00		NOCSA	1.06960%	10,704	1,538
17	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	17Y	1.00000%	6ML	0.00		NOCSA	1.10207%	16,568	1,623
18	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	18Y	1.00000%	6ML	0.00		NOCSA	1.12836%	21,911	1,707
19	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	19Y	1.00000%	6ML	0.00		NOCSA	1.14941%	26,740	1,790
20	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	20Y	1.00000%	6ML	0.00		NOCSA	1.16600%	31,069	1,872

Term	Rate	+/- Bps
2Y	-0.13300%	0.000
3Y	-0.08700%	0.000
4Y	-0.00800%	0.000
5Y	0.09300%	0.000
6Y	0.20200%	0.000
7Y	0.32100%	0.000
8Y	0.44400%	0.000
9Y	0.56100%	0.000
10Y	0.66700%	0.000
11Y	0.76300%	0.000
12Y	0.84700%	0.000
15Y	1.03000%	0.000
20Y	1.16600%	0.000
25Y	1.20100%	-
30Y	1.20400%	-
40Y	1.18600%	-
50Y	1.12000%	-

Swap Pricing, Non-Native CSA

EUR Swaps

PV

-311,136

EUR Calibration Instruments

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	Float First	CSA	Par Rate	Present Value	PV01
1	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	1Y	1.00000%	6ML	0.00		GBPCSA	-0.13970%	-11,442	100
2	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	2Y	1.00000%	6ML	0.00		GBPCSA	-0.13305%	-22,770	201
3	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	3Y	1.00000%	6ML	0.00		GBPCSA	-0.08711%	-32,855	302
4	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	4Y	1.00000%	6ML	0.00		GBPCSA	-0.00825%	-40,589	403
5	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	5Y	1.00000%	6ML	0.00		GBPCSA	0.09252%	-45,606	503
6	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	6Y	1.00000%	6ML	0.00		GBPCSA	0.20131%	-48,105	602
7	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	7Y	1.00000%	6ML	0.00		GBPCSA	0.32012%	-47,675	701
8	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	8Y	1.00000%	6ML	0.00		GBPCSA	0.44295%	-44,513	799
9	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	9Y	1.00000%	6ML	0.00		GBPCSA	0.55990%	-39,433	896
10	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	10Y	1.00000%	6ML	0.00		GBPCSA	0.66600%	-33,103	991
11	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	11Y	1.00000%	6ML	0.00		GBPCSA	0.76222%	-25,802	1,085
12	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	12Y	1.00000%	6ML	0.00		GBPCSA	0.84647%	-18,081	1,178
13	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	13Y	1.00000%	6ML	0.00		GBPCSA	0.91748%	-10,470	1,269
14	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	14Y	1.00000%	6ML	0.00		GBPCSA	0.97836%	-2,941	1,359
15	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	15Y	1.00000%	6ML	0.00		GBPCSA	1.03030%	4,386	1,447
16	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	16Y	1.00000%	6ML	0.00		GBPCSA	1.07020%	10,769	1,534
17	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	17Y	1.00000%	6ML	0.00		GBPCSA	1.10295%	16,680	1,620
18	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	18Y	1.00000%	6ML	0.00		GBPCSA	1.12949%	22,079	1,705
19	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	19Y	1.00000%	6ML	0.00		GBPCSA	1.15076%	26,971	1,789
20	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	20Y	1.00000%	6ML	0.00		GBPCSA	1.16753%	31,367	1,872

Term	Rate	+/- Bps
2Y	-0.13300%	-0.005
3Y	-0.08700%	-0.011
4Y	-0.00800%	-0.025
5Y	0.09300%	-0.048
6Y	0.20200%	-0.069
7Y	0.32100%	-0.088
8Y	0.44400%	-0.105
9Y	0.56100%	-0.110
10Y	0.66700%	-0.100
11Y	0.76300%	-0.078
12Y	0.84700%	-0.053
15Y	1.03000%	0.030
20Y	1.16600%	0.153
25Y	1.20100%	-
30Y	1.20400%	-
40Y	1.18600%	-
50Y	1.12000%	-
		-

The End